UGANDA REVISION HUB

S1 Starter Study Notes

Batch 1 • Online and offline revision

S1 • Mathematics

Integers and the number line

Learning objectives

- Recognise positive and negative integers
- Compare integers using a number line
- Add integers with the same sign

Lesson notes

Integers are whole numbers, their negatives, and zero. Examples are -4, -1, 0, 2 and 9. Fractions and decimals such as 1/2 and 2.4 are not integers.

On a number line, numbers increase as you move right and decrease as you move left. Therefore -2 is greater than -5 because -2 lies to the right of -5.

When adding integers with the same sign, add their sizes and keep the sign. For example, -3 + -2 = -5. When signs differ, subtract the smaller absolute value from the larger and keep the sign of the number with the larger absolute value.

Worked example

-6 + 2 = -4 because 6 has the larger absolute value and its sign is negative.

Revision questions

1. Arrange -3, 4, 0 and -1 in ascending order.
2. Calculate -5 + -7.
3. Calculate 8 + (-11).

Suggested answers

1. -3, -1, 0, 4
2. -12
3. -3

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S1 • Biology

Introduction to cells

Learning objectives

- Define a cell
- Name basic parts of a typical cell
- Compare plant and animal cells

Lesson notes

A cell is the basic structural and functional unit of a living organism. Some organisms consist of one cell, while plants and animals are made of many cells.

The cell membrane controls movement of substances into and out of the cell. Cytoplasm is where many chemical reactions occur. The nucleus contains genetic material and controls many cell activities.

Plant cells usually have a cellulose cell wall, chloroplasts in photosynthetic tissues, and a large permanent vacuole. Animal cells do not have a cellulose cell wall or chloroplasts.

Worked example

A leaf cell may contain chloroplasts to absorb light for photosynthesis; a cheek cell does not contain chloroplasts.

Revision questions

1 What is a cell?

2 State the function of the cell membrane.

3 Give two structures found in a typical plant cell but not an animal cell.

Suggested answers

1 The basic structural and functional unit of life

2 Controls movement of substances into and out of the cell

3 Cellulose cell wall and chloroplasts (in photosynthetic cells)

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